

Locomotion and Movement

Topic - Full Chapter

1. Given below are two statements :

Statement-I: Axial skeleton comprises 126 bones distributed along the main axis of the body.

Statement-II: The skull region articulates with the superior region of the vertebral column with the help of three occipital condyles.

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct, but statement II is incorrect.
- (3) Statement I is incorrect, but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

2. Out of 'X' pairs of ribs in humans only 'Y' pairs are true ribs. Select the option that correctly represents values of X and Y and provides their explanation.

- (1) X=12, Y=7 - True ribs are attached dorsally to vertebral column and ventrally to the sternum
- (2) X=12, Y=5 - True ribs are attached dorsally to vertebral column and sterbyn on the two ends
- (3) X=24, Y=7 - True ribs are dorsally attached to vertebral column but are free on ventral side
- (4) X=24, Y=12 - True ribs are dorsally attached to vertebral column but are free on ventral side

3. For articulation of head of humerus, a depression found in scapula is called;

- (1) Acetabulum
- (2) Manubrium
- (3) Occipital condyle
- (4) Glenoid cavity

4. Ilium is a part of;

- (1) Hindlimbs
- (2) Pelvic girdle
- (3) Pectoral girdle
- (4) Forelimbs

5. Match List-I with List-II

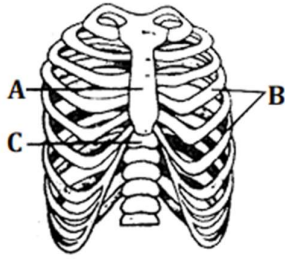
List-I	List-II
A. Sternum	i. Synovial fluid
B. Glenoid Cavity	ii. Vertebrae
C. Freely movable joint	iii. Pectoral girdle
D. Cartilagenous joint	iv. Flat bones

- (1) A-ii, B-i, C-iii, D-iv
- (2) A-iv, B-iii, C-i, D-ii
- (3) A-ii, B-i, C-iv, D-iii
- (4) A-iv, B-i, C-ii, D-iii

6. Skeletal system consists of-

- (1) Only bones
- (2) Only cartilage
- (3) A framework of bones and few cartilage
- (4) A framework of cartilage and a few bones

7. The accompanied figure is rib cage.
Identify A, B and C respectively-



- (1) Coccyx, ribs, vertebral column
- (2) Sternum, ribs, vertebral column
- (3) Scapula, ribs, vertebral column
- (4) Tarsal, ribs, vertebral column

8. Given below are two statements :

Statement-I: Many monomeric proteins called meromyosins constitute one thick filament.

Statement-II: The globular head is an active ATPase enzyme and has binding sites for ATP and active sites for actin.

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct, but statement II is incorrect.
- (3) Statement I is incorrect, but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

9. A sarcomere is best described as a :-

- (1) movable structural unit within a myofibril bounded by H zones.
- (2) fixed structural unit within a myofibril bounded By Z lines.
- (3) fixed structural unit within a myofibril bounded by A bands.
- (4) movable structural unit within a myofibril bounded by Z lines.

10. Given below are two statements :

Assertion : A motor along with muscle fibre connected to it constitute a motor unit.

Reason : The Junction between a motor neuron and the sarcolemma of the muscle fibre is called the motor end plate.

Choose the most appropriate answer from following options

- (1) Both A and R are True and R is a correct explanation of the A
- (2) Both A and R are True but R is not a correct explanation of the A
- (3) A is True but R is not False
- (4) Both A and R are False

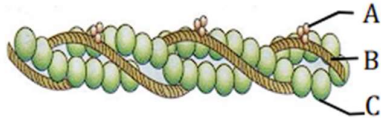
11. Which of the following statements is true?

- (1) Locomotory structures need to be different from those affecting other types of movements.
- (2) Human beings cannot move limbs, jaws, eyelids, tongue, etc.
- (3) Locomotion does not require any perfect coordinated activity.
- (4) All locomotions are movements but all movements are not locomotions.

12. Which of the following statements about human vertebral column is false?

- (1) Vertebral column consists of 26 vertebrae
- (2) It is ventrally placed
- (3) It extends from the base of skull and constitutes the main framework of the trunk
- (4) Neural canal in vertebra is the passage for spinal cord

13. See the figure of actin (thin) filaments. Identify A, B and C



- (1) A – Troponin, B – Tropomyosin, C– F-actin
- (2) A–Troponin, B–Tropomyosin, C–F-Myosin
- (3) A – Troponin, B – Myosin, C– F-Tropomyosin
- (4) A – Tropomyosin, B – Troponin, C– F-actin

14. Consider the following statements

Assertion (A): Carpals are the palm bones.

Reason (R): Wrist bones are 8 in number.

- (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is true, and Reason (R) is false.
- (4) Assertion (A) is false, and Reason (R) is true.

15. Which one of the following is incorrect?

- (1) Heart muscles are striated and involuntary.
- (2) The muscles of hands and legs are striated and voluntary.
- (3) The muscles located in the inner walls of alimentary canal are striated and involuntary.
- (4) Muscles located in the reproductive tracts are unstriated and involuntary.

16. Choose the correct option regarding a normal human :-

- A. The skull is dicondylic**
- B. Metacarpals are five in numbers**
- C. Patella is a cup-shaped bone covering the knee dorsally**
- D. Scapula is a large triangular flat bone, situated on the ventral side of the thorax**
- E. The pelvic girdle has two coxal bones**

- (1) A and E are wrong
- (2) A and B are wrong
- (3) B and E are wrong
- (4) C and D are wrong

17. Match List-I with List-II

List-I	List-II
I. Cranium / Brainbox	A. 4
II. False ribs	B. 1
III. Floating ribs	C. 8
IV. Sternum	D. 6
(1) I-A, II-B, III-D, IV-C	
(2) I-C, II-D, III-B, IV-A	
(3) I-B, II-A, III-D, IV-C	
(4) I-C, II-D, III-A, IV-B	

18. Which of the following statement is correct regarding joints?

- (1) Joints are essential for all types of movements involving bony parts of body.
- (2) Joints are points of contact between bones, or between bones and cartilages.
- (3) Force generated by the muscles is used to carry out movement through joints.
- (4) All of these

19. Given below are two statements :

Assertion (A): Osteoporosis is characterised by increased chances of fractures.

Reason (R): Decreased levels of estrogen is a common cause of osteoporosis.

(1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).

(3) Assertion (A) is true, and Reason (R) is false.

(4) Assertion (A) is false, and Reason (R) is true.

20. Identify the ribs

a. Ribs are attached to the sternum ventrally and to the vertebrae dorsally

b. Ribs are attached to sternum through costal cartilage (hyaline) of 7th rib

c. Ribs are not attached to sternum

I. True ribs

II. False ribs

III. Floating ribs

(1) a-I, b-II, c-III

(2) a-I, b-II, c-I

(3) a-II, b-I, c-III

(4) a-II, b-II, c-I

ANSWER KEY

1. (4) 2. (1) 3. (4) 4. (2) 5. (2) 6. (3) 7. (2) 8. (1) 9. (4) 10. (2)
11. (4) 12. (2) 13. (1) 14. (4) 15. (3) 16. (4) 17. (4) 18. (4) 19. (1) 20. (1)

